

# Web Application Firewall

## Home Lab using SafeLine WAF

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*VMware Workstation Pro Edition • Complete Step-by-Step Implementation Guide*

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|                         |                                |
|-------------------------|--------------------------------|
| Virtualization Platform | VMware Workstation Pro         |
| Target OS (Server)      | Ubuntu Server 22.04 LTS        |
| Attack Machine          | Kali Linux (Latest)            |
| WAF Solution            | SafeLine WAF (Community / Pro) |
| Vulnerable App          | DVWA (Damn Vulnerable Web App) |
| Difficulty Level        | Beginner — Intermediate        |

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# 1. Introduction & Lab Overview

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This guide walks you through building a fully functional cybersecurity home lab from scratch. You will deploy a vulnerable web application behind SafeLine WAF, then simulate real-world attacks to observe the firewall's detection and blocking capabilities.

This version of the guide uses VMware Workstation Pro as the hypervisor instead of VirtualBox, providing better performance, more advanced networking controls, and greater snapshot reliability — ideal for cybersecurity home labs.

## What You Will Build

The lab consists of two virtual machines on the same bridged network:

- Ubuntu Server 22.04 LTS — hosts the LAMP stack, DVWA, and SafeLine WAF
- Kali Linux — acts as the attacker machine running SQL injection and flood tests

## Lab Goals

- Set up DVWA (a deliberately vulnerable PHP application) on Ubuntu Server
- Deploy SafeLine WAF as a reverse proxy in front of DVWA
- Demonstrate SQL injection attacks and observe WAF blocking in real time
- Configure advanced WAF policies: HTTP flood defense, auth gateway, and IP deny rules

## Architecture Diagram

Traffic flow in this lab:

```
[Kali Linux Attacker]
|
| HTTP/HTTPS requests (port 443)
v
[SafeLine WAF] <-- Nginx reverse proxy, port 9443 management UI
|
| Proxied traffic (port 8080)
v
[Ubuntu Server - DVWA] <-- Apache on port 8080, MySQL backend
```

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## 2. Prerequisites

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### Host Machine Requirements

| Component  | Minimum Requirement                                                   |
|------------|-----------------------------------------------------------------------|
| RAM        | 8 GB (16 GB recommended for smooth dual-VM operation)                 |
| Disk Space | 50 GB free (SSD strongly recommended)                                 |
| CPU        | Quad-core with Intel VT-x / AMD-V enabled in BIOS                     |
| OS         | Windows 10/11, macOS 12+, or Linux (Ubuntu / Debian)                  |
| Hypervisor | VMware Workstation Pro 16 or later (free for personal use since 2024) |
| Internet   | Required for software downloads and SafeLine installation             |

### Software Downloads (All Free)

- VMware Workstation Pro — <https://www.vmware.com/products/workstation-pro.html>
- Kali Linux ISO — <https://www.kali.org/get-kali> (64-bit ISO)
- Ubuntu Server 22.04 LTS ISO — <https://ubuntu.com/download/server>
- SafeLine WAF — auto-installed via script (Section 7)

#### Good to Know

VMware Workstation Pro has been free for personal use since May 2024. Download it from Broadcom's website at <https://www.vmware.com>. You will need to create a free Broadcom account.

## 3. Lab Environment Setup — VMware Workstation Pro

### 3.1. Install VMware Workstation Pro

1. Download VMware Workstation Pro from <https://www.vmware.com/products/workstation-pro.html>
2. Run the installer. On Windows: accept UAC prompt, follow wizard. On Linux: run `chmod +x VMware-*.bundle && sudo ./VMware-*.bundle`
3. Launch VMware Workstation Pro. If prompted for a license key, select the free personal use option.
4. Enable hardware virtualization in your BIOS if VMs fail to start (look for Intel VT-x or AMD-V setting).

### 3.2. Create the Kali Linux Virtual Machine

#### Step 1 — New VM Wizard

5. Go to File > New Virtual Machine (or press Ctrl+N).
6. Select Typical (recommended) and click Next.
7. Choose Installer disc image file (iso), browse to your downloaded Kali Linux ISO, click Next.
8. Guest OS: Linux. Version: Debian 11.x 64-bit (closest match for Kali). Click Next.

## Step 2 — VM Configuration

```
Virtual Machine Name : KaliLinux
Location              : <your preferred path>
Disk Size             : 25 GB   (store as single file for performance)
Memory               : 2048 MB (2 GB)
Processors            : 2 cores
Network Adapter       : Bridged (see Section 3.4)
```

9. Click Finish. VMware will create the VM.
10. Power on the VM. Follow the Kali Linux graphical installation wizard.
11. Set hostname to kali, create a user (e.g., kali / kali).
12. After installation completes, remove the ISO: VM > Settings > CD/DVD > Use physical drive or disconnect.

## 3.3. Create the Ubuntu Server Virtual Machine

### Step 1 — New VM Wizard

13. Go to File > New Virtual Machine.
14. Select Typical, click Next.
15. Choose Installer disc image file (iso), browse to Ubuntu Server 22.04 LTS ISO.
16. VMware may auto-detect Ubuntu and offer Easy Install — skip this by selecting I will install the OS later on the next screen if you want manual control, or fill in the Easy Install fields (Full Name, Username, Password).
17. Guest OS: Linux. Version: Ubuntu 64-bit. Click Next.

### Step 2 — VM Configuration

```
Virtual Machine Name : UbuntuServer
Location              : <your preferred path>
Disk Size             : 25 GB   (store as single file)
Memory               : 4096 MB (4 GB recommended — WAF needs
headroom)
Processors            : 2 cores
Network Adapter       : Bridged (see Section 3.4)
```

18. Click Finish.
19. Power on the VM. Follow Ubuntu Server installation steps: accept defaults for partitioning, set hostname to ubuntu-server, create user ubuntu with password ubuntu (or your preferred credentials).

20. When prompted, tick Install OpenSSH server — this lets you SSH in from your host machine for easier copy-paste of commands.
21. Complete the installation and reboot.

### 3.4. Configure Bridged Networking in VMware

Bridged networking makes each VM appear as a real device on your physical LAN, allowing the VMs to communicate with each other and the host using real IP addresses.

#### Set Bridged Adapter on Both VMs

22. With the VM powered off, go to VM > Settings > Network Adapter.
23. Select Bridged: Connected directly to the physical network.
24. Check Replicate physical network connection state if you use Wi-Fi.
25. Click OK and power on the VM.

#### VMware Virtual Network Editor (if needed)

If automatic bridging doesn't work, open Edit > Virtual Network Editor (Run as Administrator on Windows):

26. Select VMnet0 (the bridged network).
27. Change Bridged to: your active network adapter (Ethernet or Wi-Fi).
28. Click Apply.

#### Find VM IP Addresses

On Ubuntu Server, after installation (Section 4.1 installs net-tools):

```
ip addr show
# or after installing net-tools:
ifconfig
```

On Kali Linux:

```
ip addr show
ifconfig
```

#### ⚠ Important

Write down both IP addresses. You will use <Ubuntu IP> throughout this guide.

If the VMs cannot ping each other, ensure both are set to Bridged and your host firewall is not blocking ICMP.

### 3.5. Take a Snapshot Before You Begin (Recommended)

VMware snapshots let you roll back instantly if something breaks — a huge advantage over VirtualBox for home labs.

29. With the VM running and base OS installed, go to VM > Snapshot > Take Snapshot.

30. Name it: Base Install — Clean State.

31. Click Take Snapshot.

Repeat for both VMs. You can always revert with VM > Snapshot > Revert to Snapshot.

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## 4. Ubuntu Server Configuration

### 4.1. Initial System Updates & Tool Installation

SSH into your Ubuntu Server from your host terminal, or type directly in the VMware console:

```
# Update package index and upgrade all packages
sudo apt-get update
sudo apt-get upgrade -y

# Install essential networking and SSL tools
sudo apt-get install -y net-tools openssl curl git
```

### 4.2. Install and Configure the LAMP Stack

LAMP (Linux, Apache, MySQL, PHP) provides the web server and database that DVWA runs on.

```
# Install Apache, PHP, MySQL in one command
sudo apt-get install -y apache2 php php-mysql mysql-server

# Verify Apache is running
sudo systemctl status apache2

# Secure MySQL (recommended)
sudo mysql_secure_installation
# Follow prompts: set root password, remove anonymous users,
# disallow remote root login, remove test database.
```

### 4.3. Install and Configure DVWA

#### Clone the DVWA Repository

```
cd /var/www/html
sudo git clone https://github.com/digininja/DVWA.git
```

#### Set Correct File Permissions

```
sudo chown -R www-data:www-data /var/www/html/DVWA
```

```
sudo chmod -R 755 /var/www/html/DVWA
```

## Configure DVWA Database Credentials

Copy the sample config and edit it:

```
cd /var/www/html/DVWA/config
sudo cp config.inc.php.dist config.inc.php
sudo nano config.inc.php
```

Update the following values in the config file:

```
$_DVWA[ 'db_database' ] = 'dvwa';
$_DVWA[ 'db_user' ]     = 'dvwa_user';
$_DVWA[ 'db_password' ] = 'p@ssw0rd';
$_DVWA[ 'db_server' ]   = '127.0.0.1';
```

## Create the MySQL Database and User

```
sudo mysql -u root -p

-- Inside MySQL shell:
CREATE DATABASE dvwa;
CREATE USER 'dvwa_user'@'localhost' IDENTIFIED BY 'p@ssw0rd';
GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

## Enable Required PHP Extensions

```
sudo apt-get install -y php-gd php-xml
sudo phpenmod gd
sudo systemctl restart apache2
```

## Initialise DVWA via Browser

32. From Kali (or your host), open a browser and visit: <http://<Ubuntu IP>/DVWA/setup.php>
33. Click Create / Reset Database.
34. You should see green success messages. If you see errors, re-check config.inc.php values.
35. Log in at <http://<Ubuntu IP>/DVWA/login.php> using default credentials: admin / password

## 4.4. Change Apache Listening Port to 8080

SafeLine WAF will listen on port 443 (HTTPS) and proxy traffic to DVWA on port 8080. This keeps the backend separate from the WAF frontend.

```
# Edit the Apache port configuration
sudo nano /etc/apache2/ports.conf

# Change: Listen 80
```

```
# To:      Listen 8080
# Edit the default virtual host
sudo nano /etc/apache2/sites-available/000-default.conf

# Change:  <VirtualHost *:80>
# To:      <VirtualHost *:8080>
# Restart Apache to apply changes
sudo systemctl restart apache2

# Verify DVWA is now on port 8080
curl -I http://localhost:8080/DVWA/
```

## 4.5. Add Sample Data to DVWA Database

Add custom user records to the database so SQL injection attacks have something interesting to return:

```
sudo mysql -u root -p

USE dvwa;

CREATE TABLE test_users (
  id      INT      NOT NULL AUTO_INCREMENT,
  username VARCHAR(50) NOT NULL,
  password VARCHAR(50) NOT NULL,
  role    VARCHAR(20) DEFAULT 'user',
  PRIMARY KEY (id)
);

INSERT INTO test_users (username, password, role) VALUES
  ('alice', 'alice123', 'user'),
  ('bob',   'bob456',   'user'),
  ('admin', 'admin123', 'admin'),
  ('charlie', 'charlie789', 'user');

EXIT;
```

---

## 5. DNS Resolution Setup

### 5.1. Quick Method — /etc/hosts

The simplest approach: edit the hosts file on both machines so the domain name dvwa.local resolves to your Ubuntu Server's IP.

#### On Ubuntu Server

```
sudo nano /etc/hosts

# Add this line (replace with your actual Ubuntu IP):
<Ubuntu IP>    dvwa.local    www.dvwa.local
```

## On Kali Linux

```
sudo nano /etc/hosts

# Add the same line:
<Ubuntu IP>      dvwa.local      www.dvwa.local
```

Test from Kali:

```
ping dvwa.local
# Should resolve to <Ubuntu IP>
```

## 5.2. Optional — BIND9 DNS Server

For a more realistic lab with a proper DNS server, install BIND9 on Ubuntu Server:

```
sudo apt-get install -y bind9
```

### Configure Zone File

```
# Edit named.conf.local
sudo nano /etc/bind/named.conf.local

zone "dvwa.local" {
    type master;
    file "/etc/bind/zones/db.dvwa.local";
};
# Create the zone directory
sudo mkdir -p /etc/bind/zones

# Create the zone file
sudo nano /etc/bind/zones/db.dvwa.local
;
; BIND data file for dvwa.local
;
$TTL 604800
@      IN  SOA  ns.dvwa.local. admin.dvwa.local. (
                    1          ; Serial
                    604800     ; Refresh
                    86400      ; Retry
                    2419200    ; Expire
                    604800 )   ; Negative Cache TTL
;
@      IN  NS   ns.dvwa.local.
ns     IN  A    <Ubuntu IP>
www    IN  A    <Ubuntu IP>
@      IN  A    <Ubuntu IP>
sudo systemctl restart bind9

# On Kali, set DNS:
sudo nano /etc/resolv.conf
# Add: nameserver <Ubuntu IP>

# Test:
dig www.dvwa.local
```

## 6. Creating a Self-Signed SSL Certificate

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SafeLine WAF will terminate HTTPS traffic. It needs an SSL certificate to do so. For the home lab, we generate a self-signed certificate using OpenSSL.

```
# Create a dedicated directory for the certificate
sudo mkdir -p /etc/ssl/dvwa

# Generate a 2048-bit RSA key and self-signed certificate (valid 365
days)
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
    -keyout /etc/ssl/dvwa/dvwa.key \
    -out /etc/ssl/dvwa/dvwa.crt \
    -subj "/C=US/ST=Lab/L=HomeLab/O=DVWA-Lab/CN=dvwa.local"
```

Verify the certificate was created:

```
ls -lh /etc/ssl/dvwa/
# Expected output:
# -rw-r--r-- 1 root root 1.2K dvwa.crt
# -rw----- 1 root root 1.7K dvwa.key

# Inspect certificate details:
openssl x509 -in /etc/ssl/dvwa/dvwa.crt -text -noout | head -20
```

### ⚠ Self-Signed Certificate Warning

Browsers will show a security warning for self-signed certificates. This is expected in a lab environment.

In production, use a certificate from Let's Encrypt or a commercial CA instead.

When SafeLine WAF prompts you to import the certificate, use the full paths above.

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## 7. Installing and Configuring SafeLine WAF

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### 7.1. Automatic Installation

SafeLine WAF installs via a single script that sets up Docker containers for all components.

```
# Ensure Docker is available (SafeLine installs it if missing)
# Run the SafeLine installer on Ubuntu Server:
bash -c "$(curl -fsSLk
https://waf.chaitin.com/release/latest/manager.sh)" -- --en
```

Follow the on-screen prompts. The installer will:

- Install Docker and Docker Compose if not present
- Pull SafeLine WAF Docker images
- Configure Nginx reverse proxy
- Start all services

- Print your admin URL, username, and auto-generated password

```
# Example output (your values will differ):  
SafeLine Management URL : https://<Ubuntu IP>:9443  
Admin Username          : admin  
Admin Password          : xKj8mP2qR7vN (auto-generated)
```

### ⚠ Save Your Credentials

Save the admin password immediately — it is shown only once.

The management UI runs on port 9443 (HTTPS). You will access it from Kali or your host browser.

Accept the self-signed certificate warning in your browser to proceed.

## 7.2. First Login and License

36. Open a browser and navigate to: `https://<Ubuntu IP>:9443`
37. Log in with admin and the password printed by the installer.
38. You will be prompted to change your password on first login — do so.
39. The Community (free) license covers all features needed for this lab.
40. Optional: Upgrade to a 7-day Pro trial at <https://ly.safepoint.cloud/zfYZD3l> for \$1 to explore pro features.

## 7.3. Import the Self-Signed SSL Certificate

41. In the SafeLine management UI, navigate to Certificates (usually under Settings or SSL).
42. Click Add Certificate or Import.
43. Paste the contents of `/etc/ssl/dvwa/dvwa.crt` in the Certificate field.
44. Paste the contents of `/etc/ssl/dvwa/dvwa.key` in the Private Key field.

```
# To copy certificate content:  
sudo cat /etc/ssl/dvwa/dvwa.crt  
  
# To copy key content:  
sudo cat /etc/ssl/dvwa/dvwa.key
```

45. Click Save. The certificate should now appear in the Certificates list.

## 7.4. Onboard the DVWA Application

This is the core configuration — you tell SafeLine WAF to sit in front of DVWA.

46. In the SafeLine UI, go to Site or Applications and click Add Site / Add Application.
47. Fill in the fields as follows:

| Setting                | Value                                       |
|------------------------|---------------------------------------------|
| Domain / Server Name   | www.dvwa.local (or dvwa.local)              |
| Listen Port            | 443 (delete port 80 if present)             |
| Backend (Upstream) URL | http://<Ubuntu IP>:8080                     |
| SSL Certificate        | Select the certificate imported in Step 7.3 |
| Protection Mode        | Protection (blocking mode)                  |

48. Click Save / Create.

49. Update /etc/hosts on Kali so dvwa.local resolves to the Ubuntu Server IP (if not already done).

50. From Kali, browse to https://dvwa.local — you should be proxied through SafeLine to DVWA.

#### Tip

Traffic flow: Kali → port 443 → SafeLine WAF → port 8080 → DVWA.  
If you see a 502 Bad Gateway, verify DVWA is running on port 8080: curl http://localhost:8080/DVWA/

## 8. SQL Injection Demo from Kali Linux

### 8.1. Prepare DVWA for Testing

51. From Kali, browse to https://dvwa.local (accept certificate warning).

52. Log in with: admin / password

53. Go to DVWA Security tab (in the left menu).

54. Set security level to Low and click Submit.

55. Navigate to SQL Injection in the left menu.

### 8.2. Manual SQL Injection Test

With DVWA security on Low and SafeLine WAF in Protection mode:

56. In the User ID input box, type: 1

57. Click Submit — you should see normal output (user details for ID 1).

58. Now type: 1' OR '1'='1

59. Click Submit.

Expected result: SafeLine WAF detects the SQL injection payload and returns a block page instead of passing the request to DVWA.

### 8.3. SQLMap Automated Test from Kali

Kali Linux includes sqlmap, which automates SQL injection discovery:

```
# First, get a valid session cookie from DVWA:
# Log in via browser, open DevTools > Application > Cookies
# Copy the PHPSESSID and security cookie values

# Run sqlmap against DVWA's SQL injection page:
sqlmap -u
"https://dvwa.local/DVWA/vulnerabilities/sqli/?id=1&Submit=Submit" \
  --cookie="PHPSESSID=<your_session>; security=low" \
  --dbs --batch --risk=1 --level=1
```

### 8.4. Observe SafeLine WAF Logs

60. Return to the SafeLine management UI (<https://<Ubuntu IP>:9443>).
61. Navigate to Events or Attack Logs.
62. You should see blocked events for each injection attempt, including the source IP (Kali), attack type (SQL Injection), and the blocked payload.
63. Click on any event for detailed analysis including the raw HTTP request and the rule that triggered.

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## 9. SafeLine WAF — Advanced Configurations

### 9.1. HTTP Flood Defense (Rate Limiting)

Simulate a DoS attack from Kali and verify SafeLine rate-limits it.

#### Configure Rate Limiting in SafeLine

64. In the SafeLine UI, go to your DVWA site settings.
65. Find HTTP Flood Defense or Rate Limiting.
66. Set thresholds, for example: 50 requests per second from a single IP triggers a challenge or block.
67. Set penalty duration to 60 seconds and click Save.

#### Test with Apache Bench from Kali

```
# Send 500 requests, 50 concurrent:
ab -n 500 -c 50 https://dvwa.local/DVWA/

# Alternatively, use siege:
sudo apt-get install -y siege
```

```
siege -c 100 -t 30S https://dvwa.local/DVWA/
```

After the threshold is hit, subsequent requests should return a block page. Check SafeLine logs for flood events.

## 9.2. Authentication Gateway

SafeLine can require users to pass an auth challenge before reaching the backend — useful for adding login protection to any app.

68. In the SafeLine UI, go to your DVWA site configuration.
69. Enable Auth Sign-In (sometimes labelled Human Verification or Authentication Gateway).
70. Configure a username and password (separate from DVWA credentials).
71. Save the configuration.
72. From Kali, browse to <https://dvwa.local> — SafeLine should display a login prompt before forwarding to DVWA.

## 9.3. Custom IP Deny Rules — Block Kali

Create a rule that completely blocks your Kali machine's IP address.

73. Find Kali's IP address: `ip addr show` (e.g., 192.168.1.50).
74. In SafeLine UI, go to Deny Rules or IP Blacklist.
75. Click Add Rule. Set Match: Source IP = 192.168.1.50, Action: Block.
76. Save the rule.
77. From Kali, try to browse to <https://dvwa.local> — the connection should be immediately rejected.
78. Remove or disable the rule when testing is complete.

## 9.4. XSS Attack Test

DVWA also has an XSS (Cross-Site Scripting) module. Test SafeLine's detection:

79. In DVWA, navigate to XSS (Reflected) in the left menu.
80. In the Name field, enter: `<script>alert('XSS')</script>`
81. Click Submit.

SafeLine should block the request. Check the attack logs — you should see an XSS event alongside the SQL injection events from earlier.

---

# 10. Conclusion and Further Exploration

Congratulations! You have built a fully functional cybersecurity home lab demonstrating:

- VM provisioning and bridged networking in VMware Workstation Pro
- LAMP stack deployment with a deliberately vulnerable web application (DVWA)
- Reverse proxy configuration with SafeLine WAF
- SQL injection and XSS attack demonstrations
- WAF protection: SQL injection blocking, rate limiting, auth gateway, IP deny rules

## Ideas for Further Exploration

| Extension            | Description                                                          |
|----------------------|----------------------------------------------------------------------|
| OWASP Juice Shop     | More realistic vulnerable app with more attack surfaces (npm/Docker) |
| Metasploit Framework | Use Kali's Metasploit to run exploit modules against DVWA            |
| SIEM Integration     | Forward SafeLine WAF logs to a Splunk or Elastic SIEM instance       |
| IDS/IPS              | Add Suricata or Snort alongside SafeLine for network-level detection |
| Additional WAFs      | Compare SafeLine with ModSecurity, Nginx WAF, or AWS WAF             |
| HTTPS Everywhere     | Force HTTPS redirects and test HSTS headers through the WAF          |
| Custom WAF Rules     | Write custom regex rules in SafeLine to block specific payloads      |
| Container Lab        | Replace VMs with Docker containers for a lighter-weight setup        |

## 11. Troubleshooting Reference

| Problem                                        | Solution                                                                                                                                          |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| VMs cannot ping each other                     | Both VMs must be set to Bridged networking. Check VMware Virtual Network Editor — ensure VMnet0 is bridged to your active NIC.                    |
| DVWA shows 'Could not connect to the database' | Verify MySQL is running: <code>sudo systemctl status mysql</code> . Re-check <code>config.inc.php</code> credentials match what you set in MySQL. |
| Apache returns 403 Forbidden on /DVWA/         | Check file ownership: <code>sudo chown -R www-data:www-data /var/www/html/DVWA</code>                                                             |
| SafeLine shows 502 Bad Gateway                 | DVWA backend is not reachable. Verify Apache is on port 8080: <code>sudo ss -tlnp   grep 8080</code>                                              |
| Cannot access SafeLine UI at port 9443         | Check Docker is running: <code>sudo systemctl status docker</code> . Check containers: <code>sudo docker ps</code>                                |

| Problem                                  | Solution                                                                                                                           |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| SQLMap blocked immediately by WAF        | Expected behaviour. Try running sqlmap with --tamper=space2comment to see if evasion changes results.                              |
| Certificate warning in browser           | Self-signed certificates are not trusted by default. Click Advanced > Proceed to continue. Normal for lab use.                     |
| VMware Easy Install skips Ubuntu prompts | Select 'I will install the OS later' when creating the VM to control the full installation manually.                               |
| SafeLine installer fails                 | Ensure Ubuntu has internet access: ping google.com. Check Docker is not already running a conflicting service on port 443 or 9443. |

## Useful Commands Reference

```
# Check all running services:
sudo systemctl status apache2 mysql
sudo docker ps

# View Apache error log:
sudo tail -f /var/log/apache2/error.log

# View SafeLine WAF Docker logs:
sudo docker logs safeline-mgt 2>&1 | tail -50

# Test WAF is proxying correctly:
curl -k -H 'Host: dvwa.local' https://<Ubuntu IP>/DVWA/

# Restart SafeLine WAF:
cd /data/safeline && sudo docker-compose restart
```

---

*Keep your home lab isolated from production networks • Regularly update all components • Good luck!*